

A Self-Service Population Health Analytical Tool

The Client

An opportunity assessment team within a U.S.-based health insurer, recognized the need to harness the power of data for a deeper understanding of their members. The goal was to efficiently identify member cohorts and gain insights into the clinical and financial impacts of various levers to improve member outcomes.

Challenge

The client was utilizing a vast array of tools for data visualization and analysis. While each tool had its specific use, they collectively lacked the versatility offered by an all-encompassing, self-service analytics tool.

Maintenance of these disparate dashboards was highly time-consuming, and the client found themselves routinely writing SQL code to extract data for ad-hoc reports. The absence of standard methods for specific cohort data retrieval exacerbated the problem. They needed a reliable, intuitive dashboard where users could retrieve dependable data with ease.

Solution: a self-service analytics tool

In response to these challenges, we designed a robust, self-service analytical tool combined with data services. This solution empowers the client's business analysts to effectively test hypotheses, efficiently extract both raw and aggregated data, and enhance their ability to identify insights for improving member health outcomes. The tool not only saves them significant time but also amplifies their capacity to impact positive health outcomes.

With the easy-to-use GUI interface, users can delve into a level of detail that was previously impossible, examining data from a multitude of angles to drive decision-making. This ability has been instrumental in identifying and closing gaps in care, adjusting clinical program enrollment, and assessing a variety of other levers.

Successes

- ◆ **Data Integration:** Consolidated data from 10+ sources (Claims, SDOH, benefits, etc.) and transformed it to fit within a BigQuery instance
- ◆ **Query Optimization:** Developed optimized SQL queries to give users real-time access to billions of rows of data
- ◆ **Advanced Analytics:** Enabled users to perform complex analysis from a GUI interface such as tracking KPIs, comparing member cohorts and providing insights into the effectiveness of interventions
- ◆ **Developer Tools:** Custom R package to store commonly used data functions.

"This is the most professional tool that I have seen at this organization" VP Clinical Evaluation